

Burning fuses on RT7

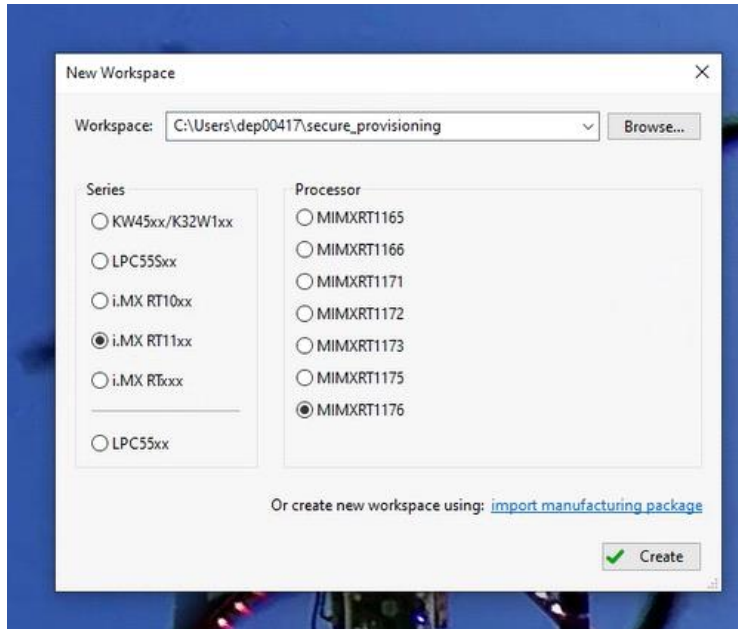
Tuesday, September 12, 2023 4:25 PM

Install NXP Mcuxpresso secure provisioning

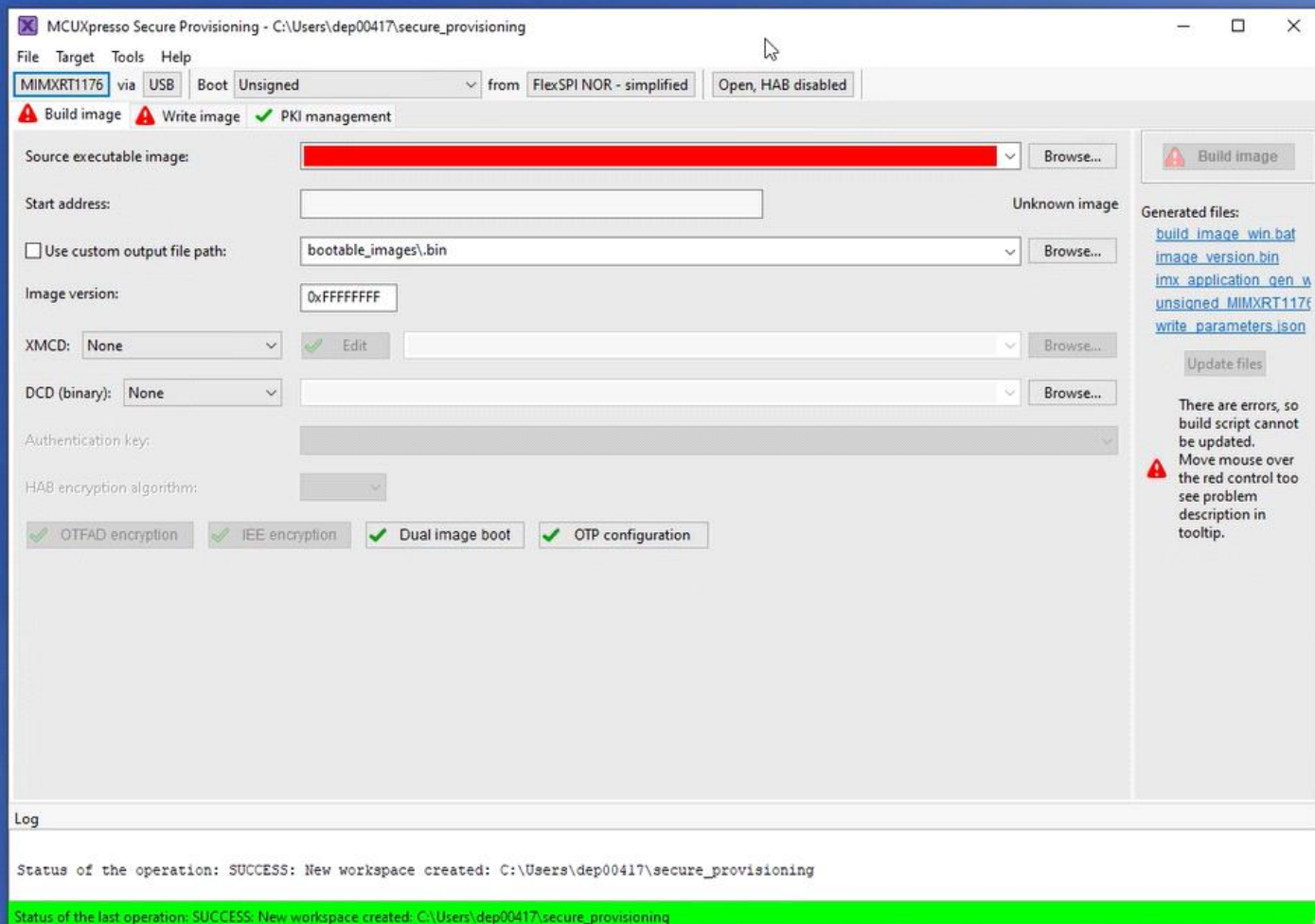
<https://www.nxp.com/design/software/development-software/mcuxpresso-software-and-tools-/mcuxpresso-secure-provisioning-tool:MCUXPRESSO-SECURE-PROVISIONING#downloads>

Start secure provisioning

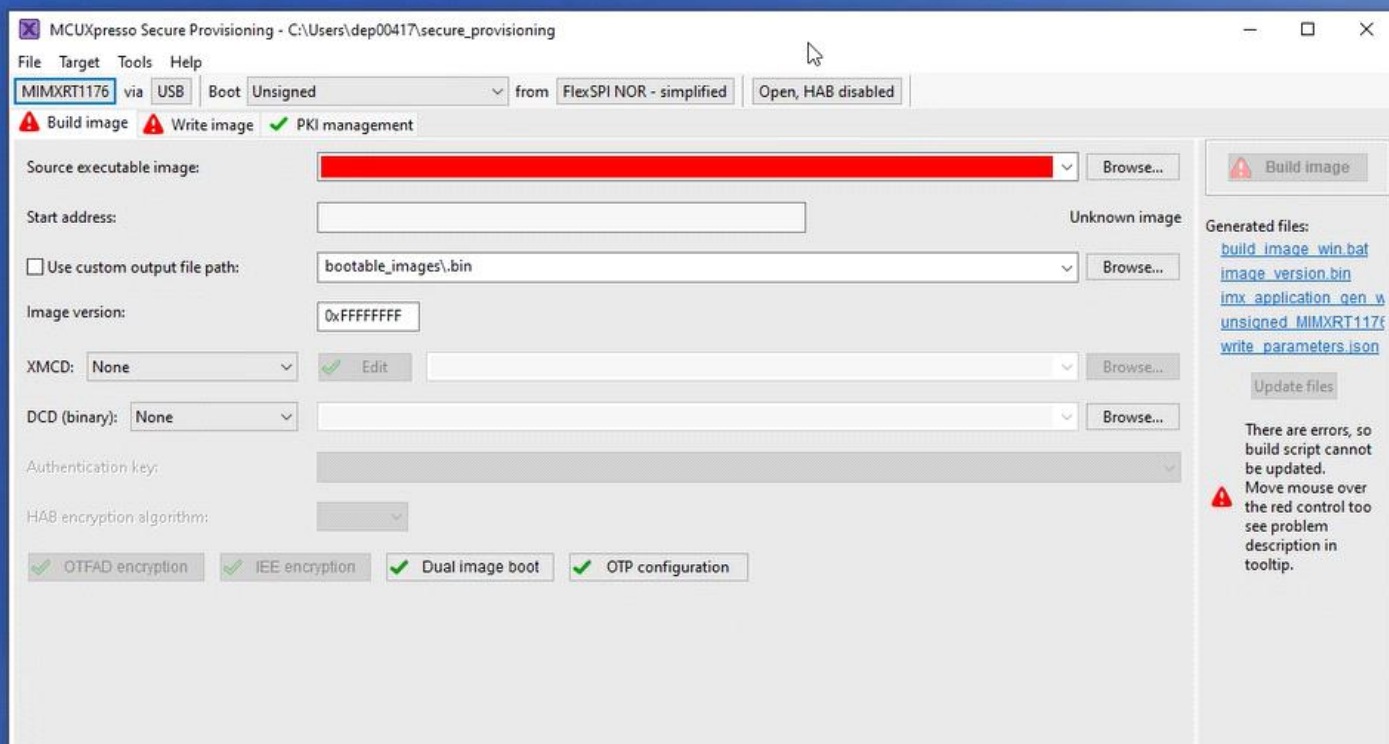
Select MIMXRT1176



Select create



Select OPT configuration



Log

Status of the operation: SUCCESS: New workspace created: C:\Users\dep00417\secure_provisioning

Status of the last operation: SUCCESS: New workspace created: C:\Users\dep00417\secure_provisioning

Press Yes

Wait for the script to finish

QTP Configuration, MIMXRT1176, USB[vid=0x1FC9,pid=0x013D]

Tree controls

Show All Search

Fuse name	Offset	#	R	W	E	Current value	Required value
Reserved 0x800-0x830							
Fuse 0x840	0x840	4				0	*
Reserved 0x850							
Fuse 0x860	0x860	6				0x0000C020	*
0x860							
Fuse 0x870	0x870	7				0x00004028	*
0x870							
Locks							
0x880	0x880	8				0x00000003	*
0x890	0x890	9				0x00000003	*
Reserved 0x8A0-0x8C0							
Fuse 0x8D0	0x8D0	13				0	*
0x8D0							
Fuse 0x8E0	0x8E0	14				0x00000202	*
0x8E0							
Reserved 0x8F0							
Fuse 0x900	0x900	16				0x82929744	*
LOT_NO_ENC[42:11]							
Fuse 0x910	0x910	17				0x0B1D180E	*
0x910							
Fuse 0x920	0x920	18				0x00003A1B	*
0x920							
Fuse 0x930	0x930	19				0x0000000A	*
0x930							
Boot config							
0x940	0x940	20				0	*
0x950	0x950	21				0	*

Value of the selected fuse

0x840	HEX
Current value	0
Required value	*

Actions

Clear Use current Fix

Locks

Write Read Lock after write

BITS	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Current value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Required value	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*

Bit-fields in the selected fuse

Bit-field	Bits	Current value	Required value
Reserved	0	Reserved/0	*
Reserved	1	Reserved/0	*
MECC_ENABLE	2	disabled/0	*
XECC_ENABLE	3	disabled/0	*
Reserved	4	Reserved/0	*
Reserved	5	Reserved/0	*
Reserved	6	Reserved/0	*
Reserved	7	Reserved/0	*
Reserved	8	Reserved/0	*
Reserved	9	Reserved/0	*
Reserved	10	Reserved/0	*
Reserved	11	Reserved/0	*
Reserved	12	Reserved/0	*
Reserved	13	Reserved/0	*

Fuse description:

Fuse 0x840

Status: No user requirements.

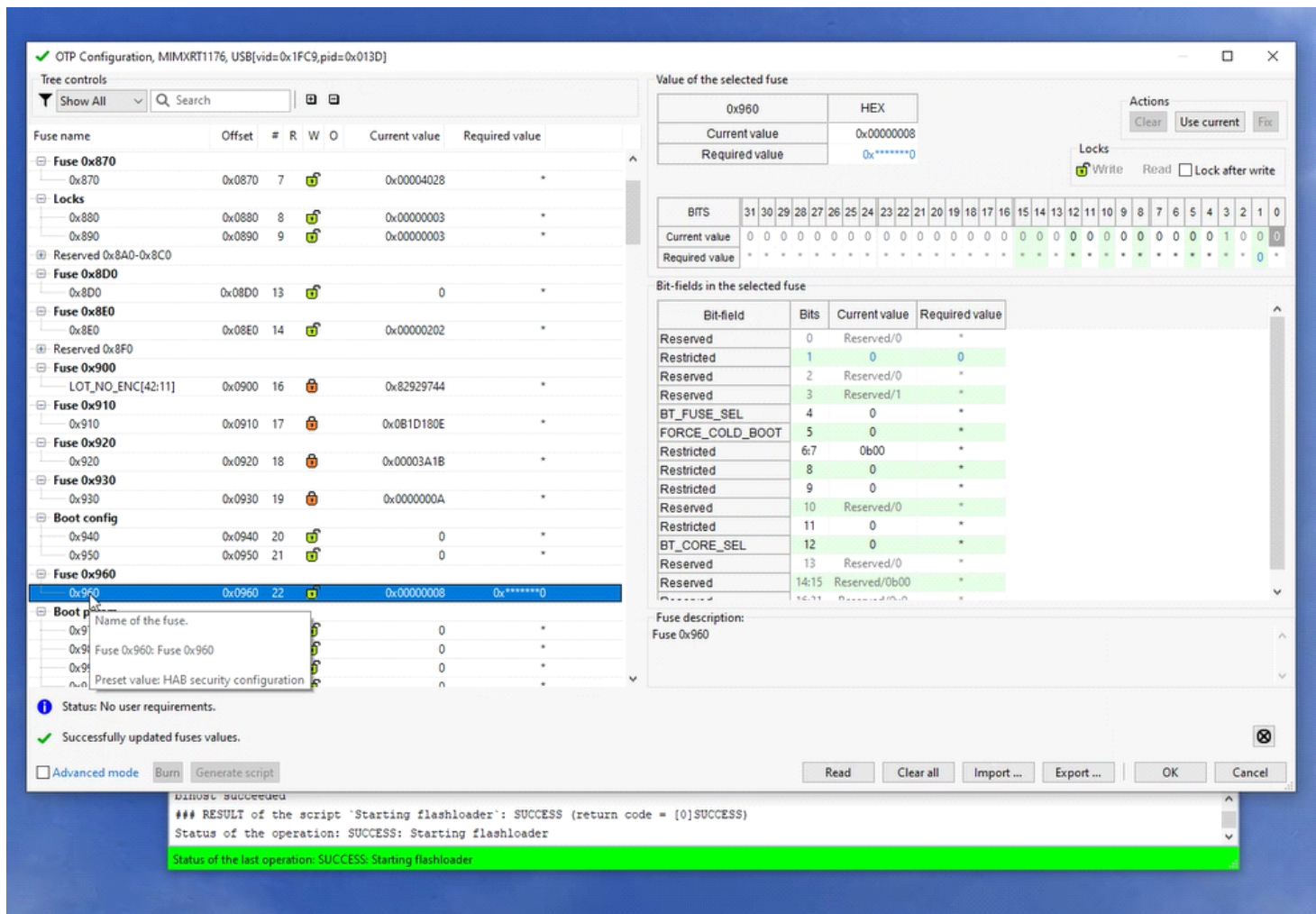
Successfully updated fuses values.

Advanced mode Burn Generate script

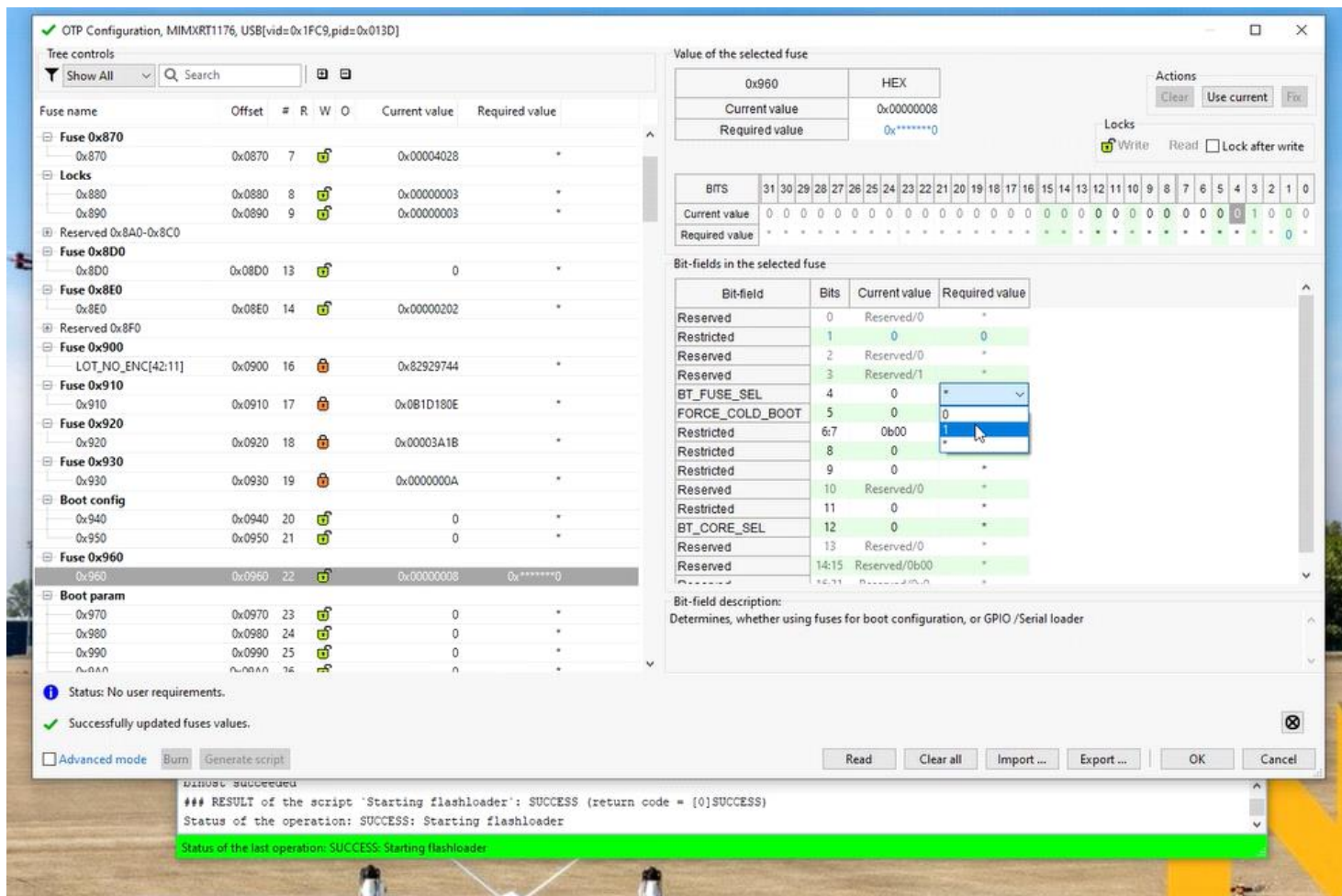
Read Clear all Import ... Export ... OK Cancel

PRG011.T of the script 'Starting flashloader': SUCCESS (return code = 10) SUCCESS!

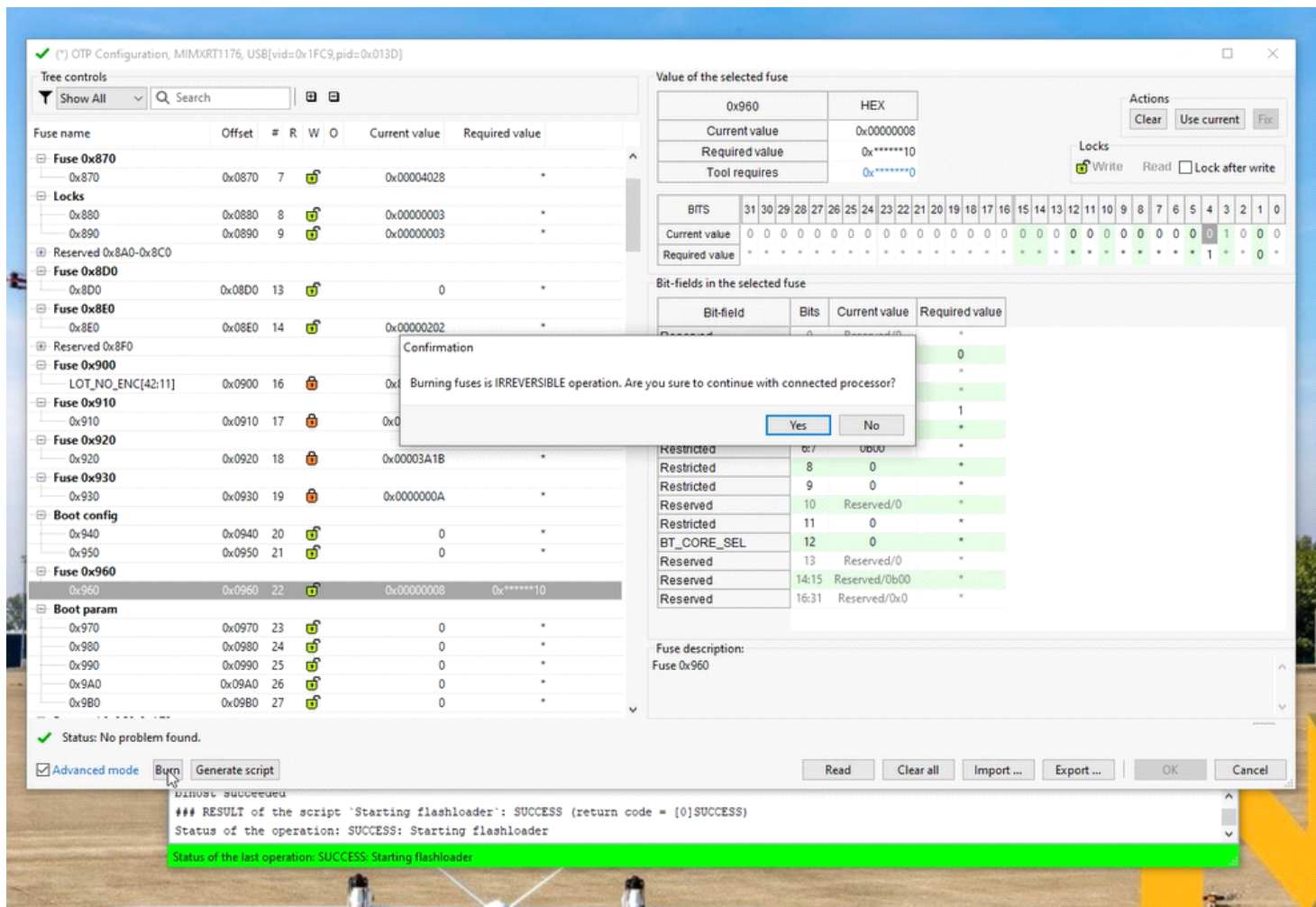
Select "0x960"
Fuse 0x960

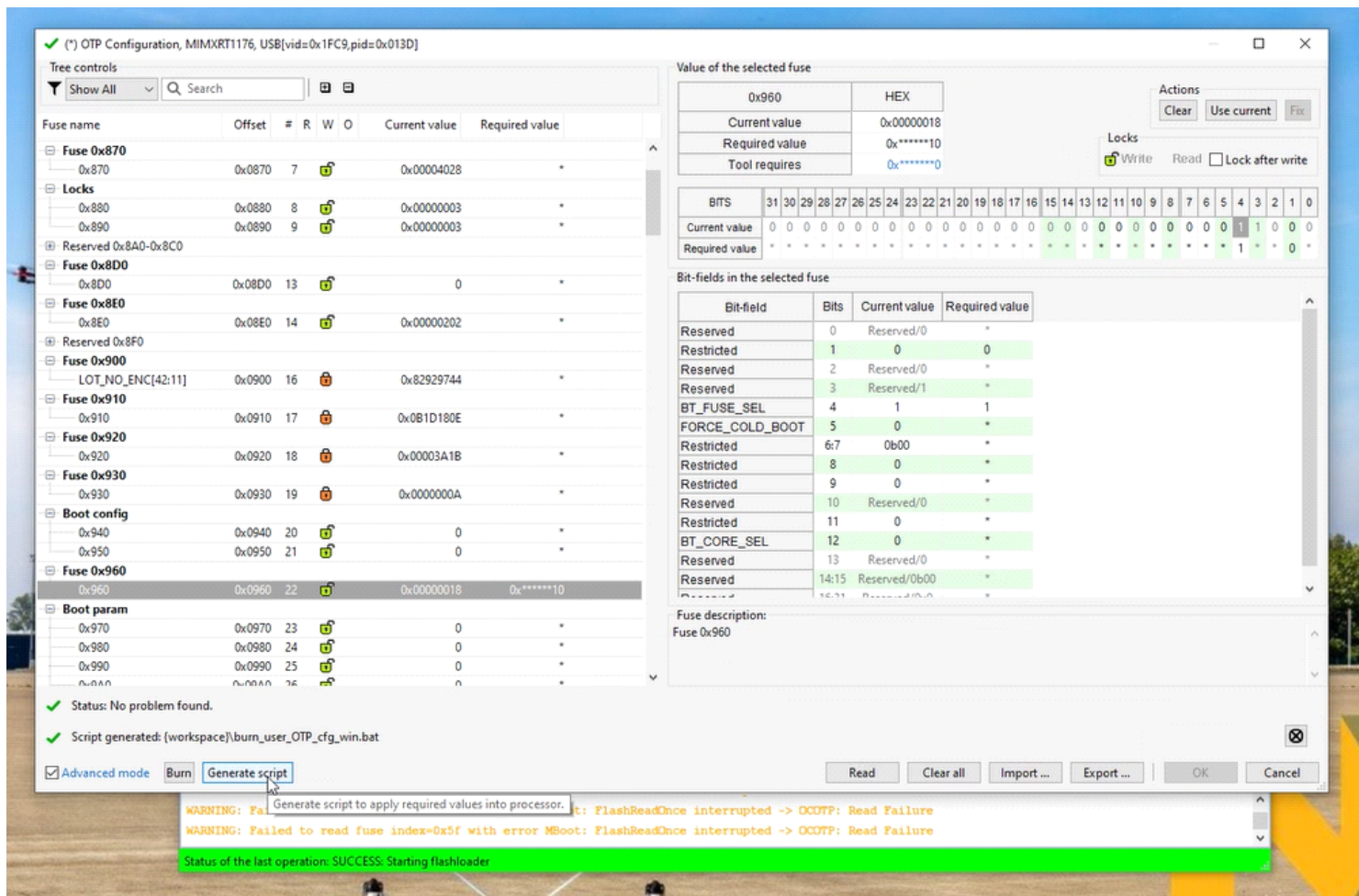


BT_FUSE_SEL select Required value 1



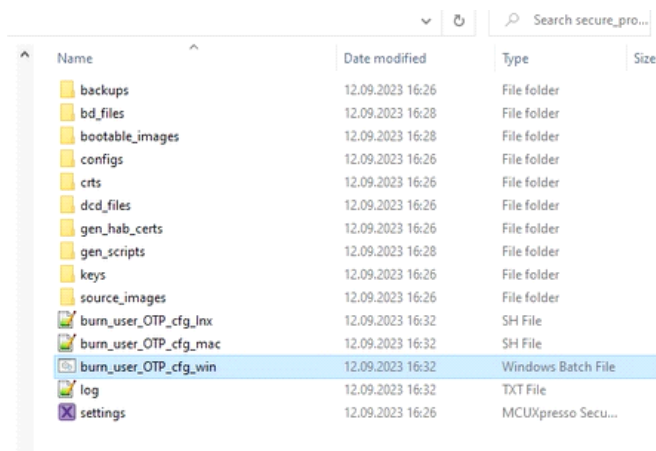
Bottom-left corner and select advanced mode





{workspace} is C \ users \ \$username \ secure provisioning

There's a burn_user_OTP_cfg_win.bat



You can use to burn fuses for new boards